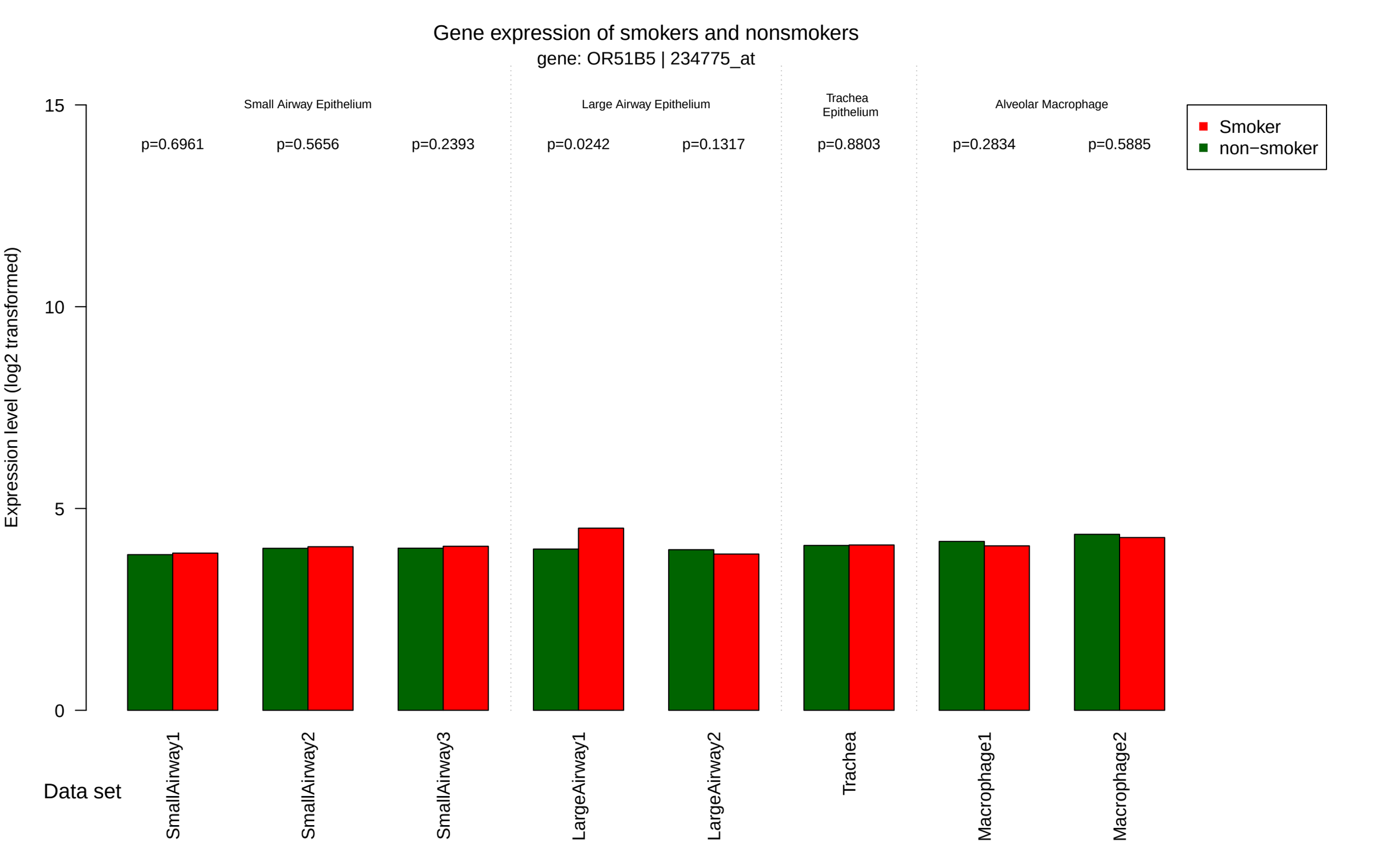
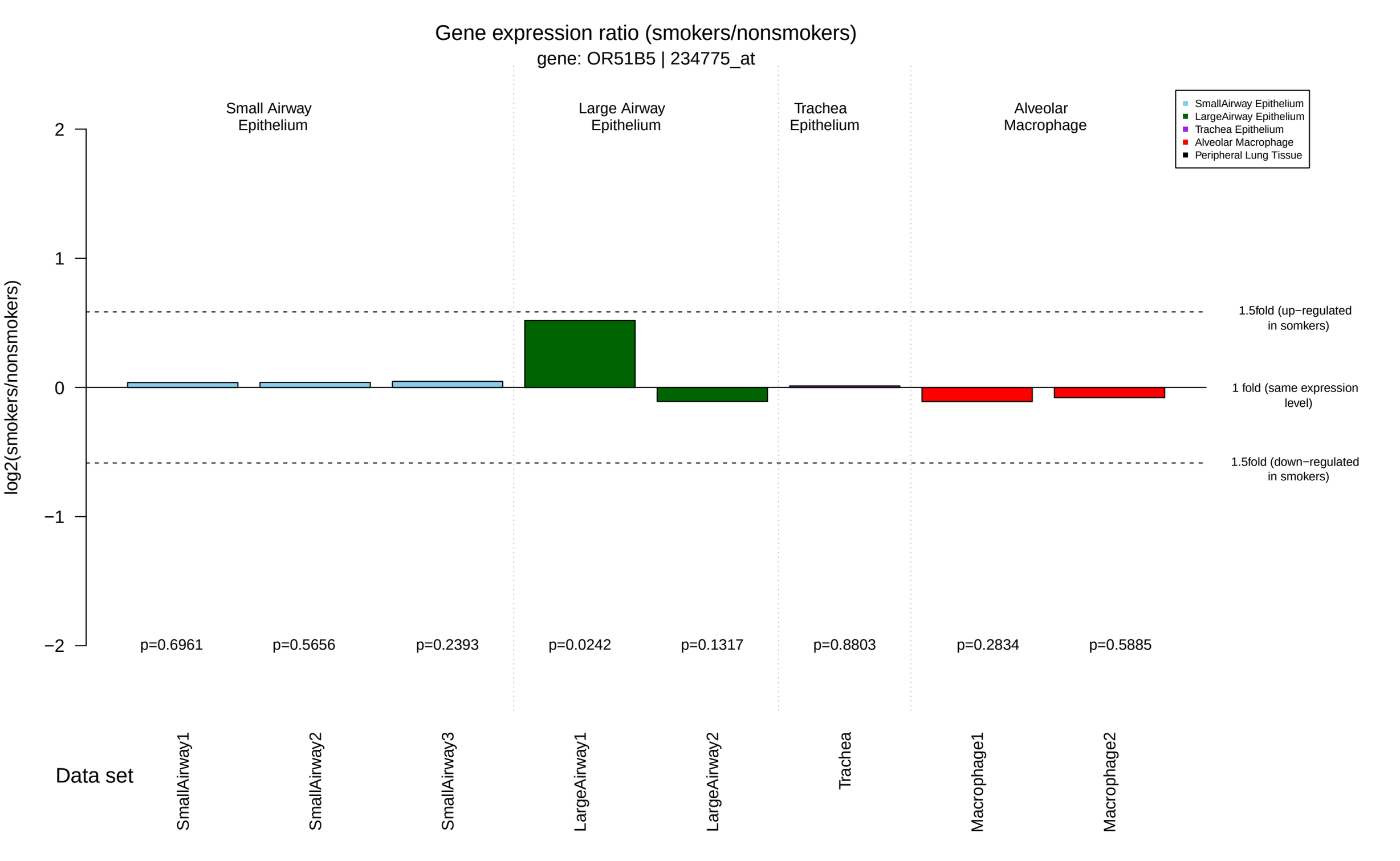
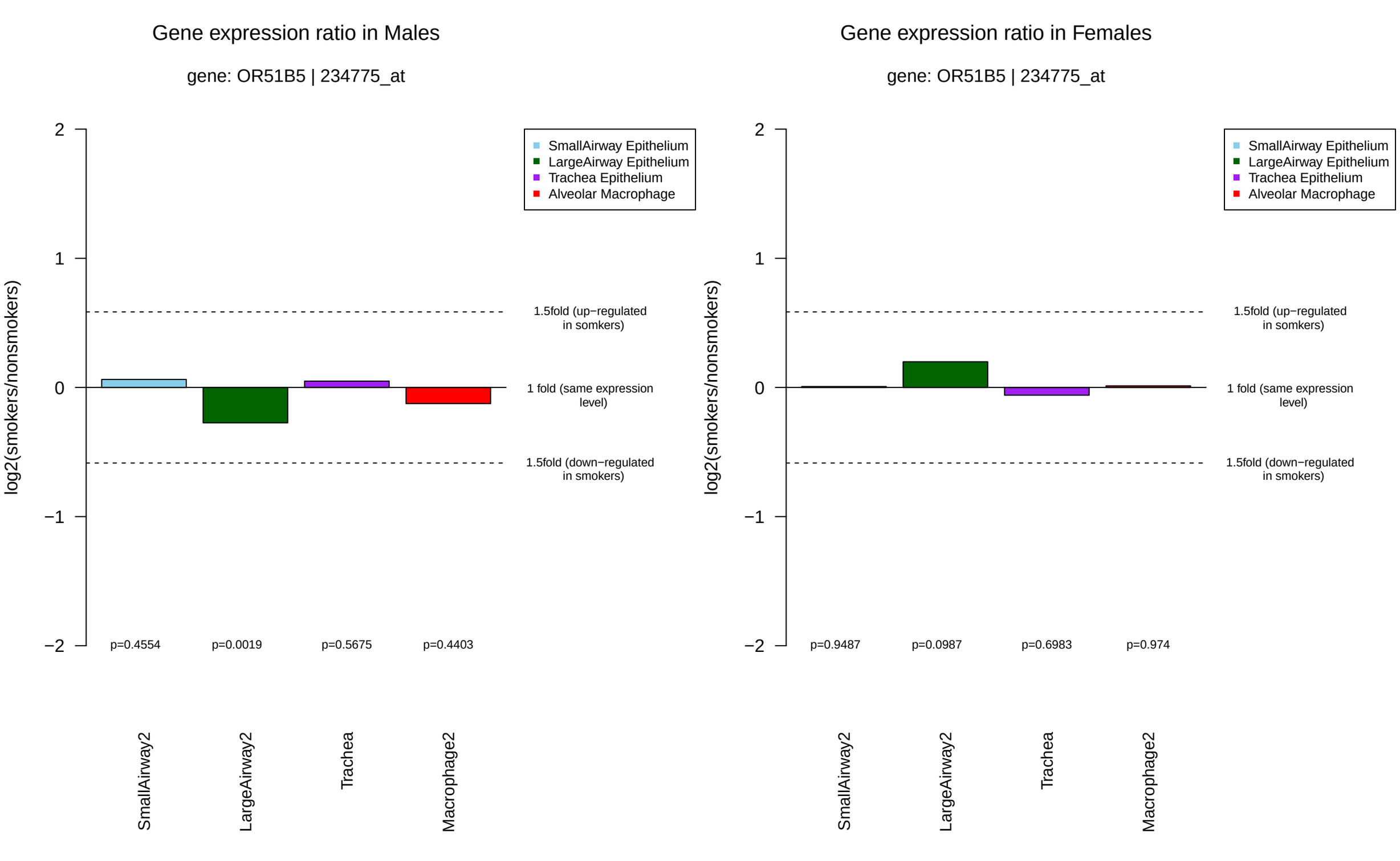
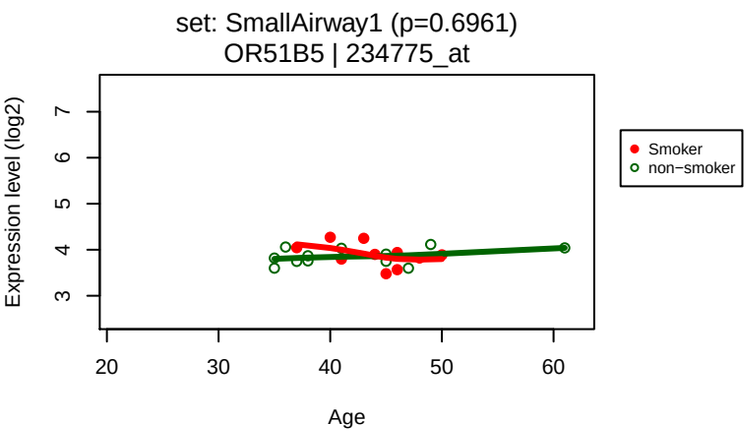
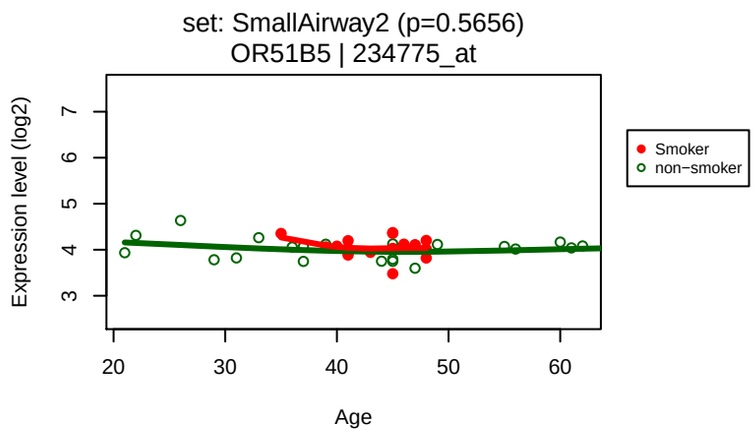
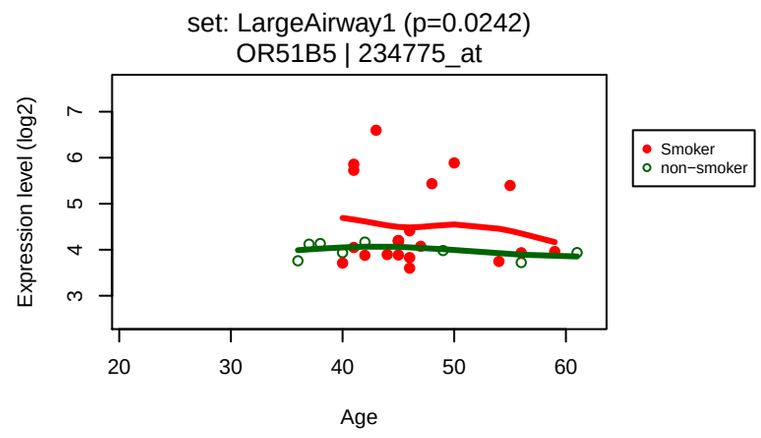

Gene expression of smokers and nonsmokers
gene: OR51B5 | 234775_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6961
p=0.5656
p=0.2393
p=0.0242
p=0.1317
p=0.8803
p=0.2834
p=0.5885
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

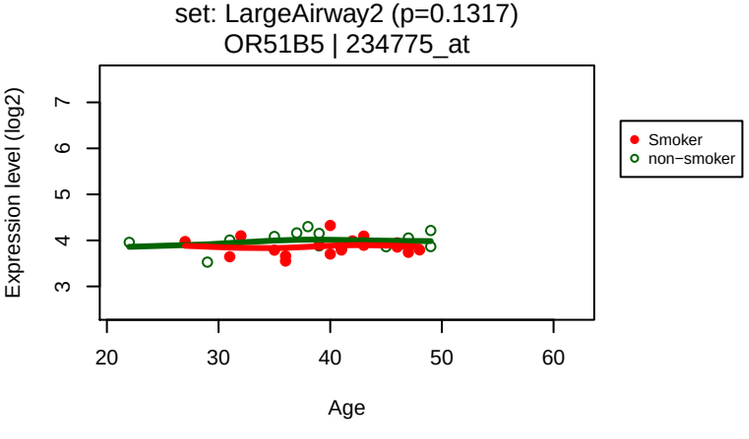

Gene expression ratio (smokers/nonsmokers)
gene: OR51B5 | 234775_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6961
p=0.5656
p=0.2393
p=0.0242
p=0.1317
p=0.8803
p=0.2834
p=0.5885
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

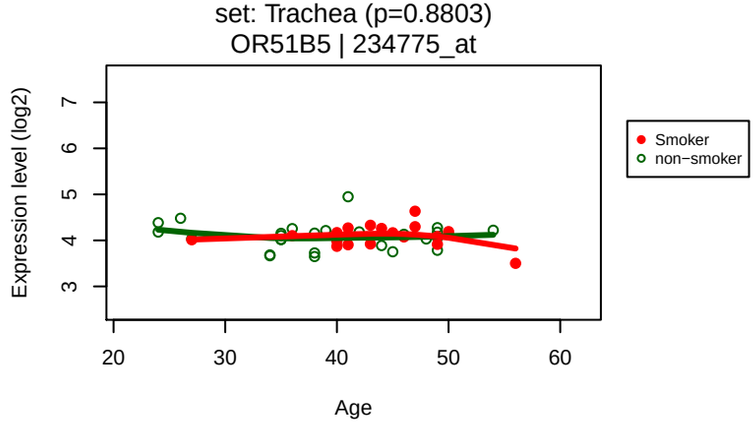

Gene expression ratio in Males
gene: OR51B5 | 234775_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4554
p=0.0019
p=0.5675
p=0.4403
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51B5 | 234775_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9487
p=0.0987
p=0.6983
p=0.974
SmallAirway2
LargeAirway2
Trachea
Macrophage2

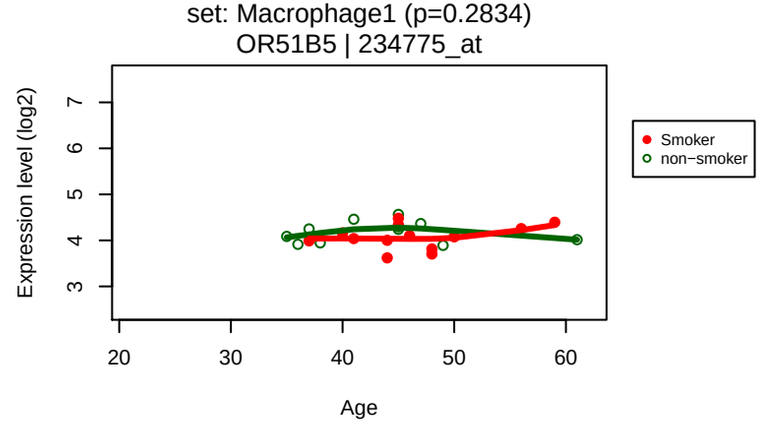

set: SmallAirway1 (p=0.6961)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker

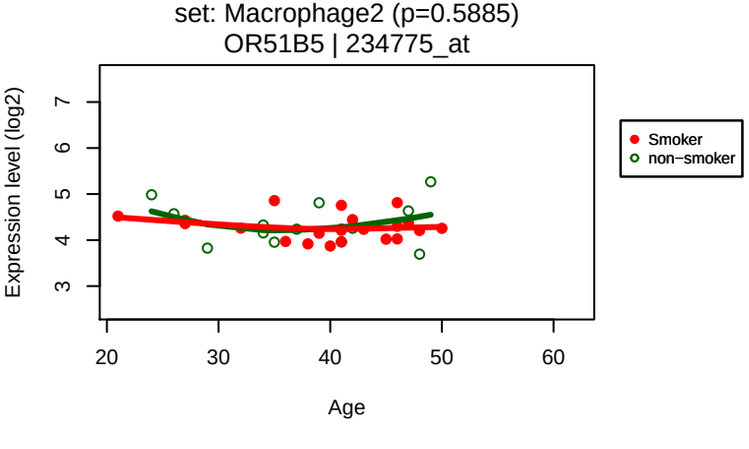

set: SmallAirway2 (p=0.5656)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker

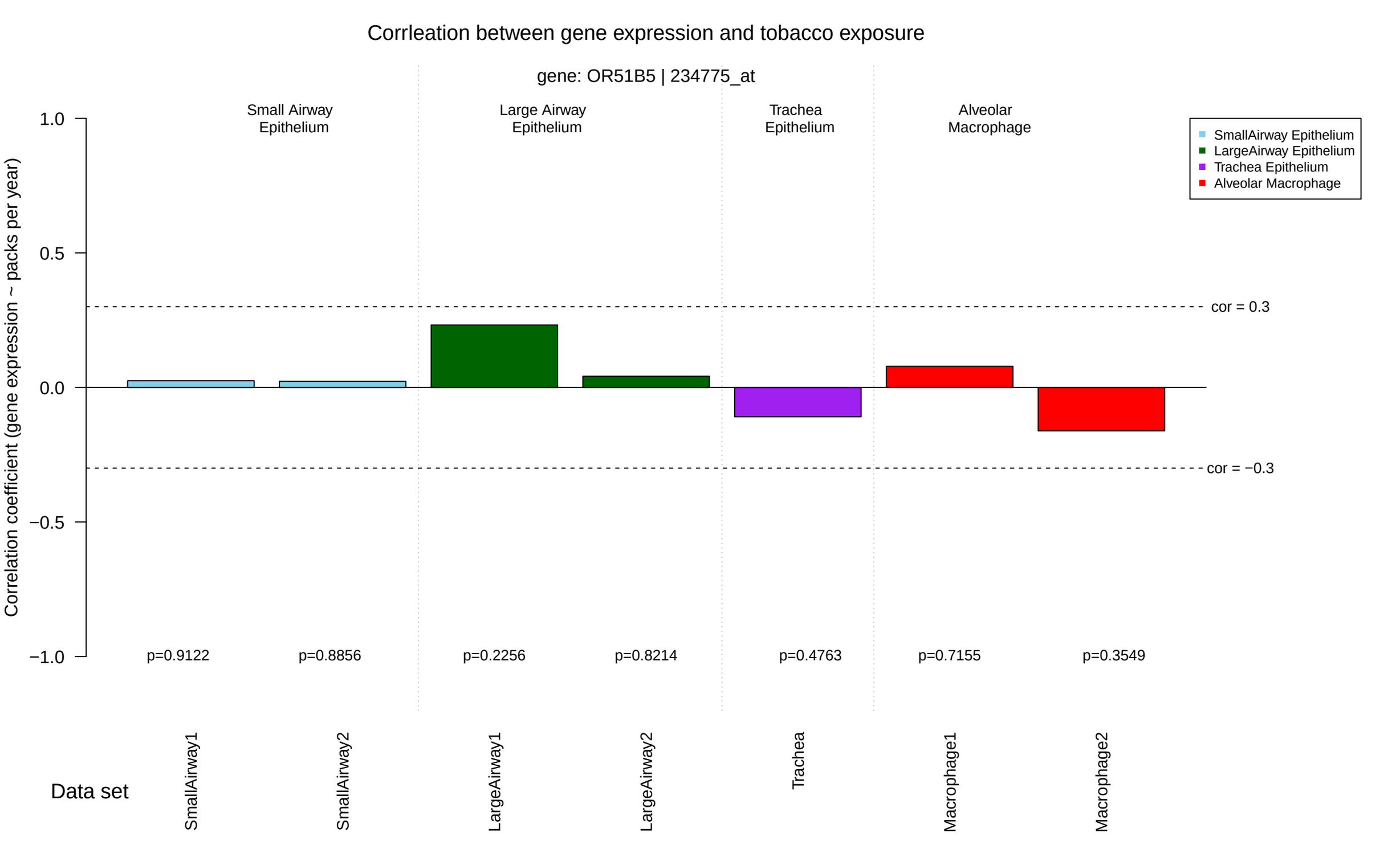

set: LargeAirway1 (p=0.0242)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1317)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8803)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2834)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5885)
OR51B5 | 234775_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51B5 | 234775_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9122
p=0.8856
p=0.2256
p=0.8214
p=0.4763
p=0.7155
p=0.3549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2